EL NIÑO ACTION BRIEFING · 30 SEPTEMBER 2026 · VERIFIED RUN

Barú, Pérez Zeledón, Costa Rica

The drought is the slow hazard and the river is the fast one. Both deaths in Barú this year were at Nauyaca Falls, from flash floods. Before the December tourist season, agree a rule that stops all river crossings when rain falls upstream.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Not resolved. Barú district (2,897 people, 2022) sits in Pérez Zeledón canton, inside a larger region that could not be confirmed	No figure published for Costa Rica or Central America	n/a (no local figure in the Sep 2026 – Feb 2027 window)	Flash floods on the Río Barú and drought / water supply; secondary: dengue, heat for outdoor workers, road isolation

19/19 recommendations checked. 1 confirmed without correction, 10 corrected, 5 unverified, 3 contraindicated. Exposure figure: Climate Impact Lab, *451,000 at Risk*, September 2026, method paper forthcoming. It publishes no figure for Costa Rica, and **nothing in this report is scaled from it.** The historical record and CNE/IMN bulletins carry the hazard instead. **A correction to the brief:** Dominical, Dominicalito, and Escaleras are not in Barú. Dominical is in Bahía Ballena district, Osa canton, which the Uvita, Osa briefing already covers. Whether Dominicalito and Escaleras sit in Osa is not yet confirmed from a primary source. Barú is the inland district of the Río

* Suggested formulation: check with a health worker before sharing.

EXECUTIVE SUMMARY — AT A GLANCE



9.27°N 83.75°W, Costa Rica · Natural Earth

- **The risk.** Flash floods on the Río Barú are the killer: both 2026 deaths were at Nauyaca Falls, a tourist in June and a guide on 30 August, during a 25% rainfall deficit.
- **First action.** Agree a flash-flood stop rule for Nauyaca and every Río Barú crossing, owned by the Pérez Zeledón emergency committee, tour operators, and ICT, by 15 October 2026.
- **The money.** $25,400 floor, $32,900 target, ~$72,700 ceiling (provisional). Top placement: a $15,000 Cruz Roja Pérez Zeledón contingency fund released if the CNE raises the canton to orange.
- **Do not.** Do not drill new wells or pump existing ones harder, because adding users to springs and sources that are already falling repeats the mistake that ruined Guanacaste's aquifers.

What to do	**10** actions	p. 2
Who does what	RACI chart	p. 9
Where money does the most	funding	p. 9
Do not do this	**4** items	p. 11
What has happened here	the record	p. 13
How this was produced	methodology	p. 15
References	sources	p. 16

Barú valley, with its district centre at Platanillo and settlements including Tinamaste, San Salvador, and Alfombra. Nauyaca Falls is in San Salvador de Barú.

WHAT THIS PLACE SHOULD DO

RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached independently is the stronger bet. With about 2,900 residents, expect roughly 15 deaths a year from all causes (an assumed crude rate, not a sourced figure), so no action's effect will show up in deaths. Track flash-flood incidents, water outage days, dengue admissions, and heat illness instead.

1. Agree a flash-flood stop rule for Nauyaca and every Río Barú crossing by 15 October

PARTIALLY CONFIRMED | EMERGENCY · HISTORIAN · ECONOMIST · FRONTLINE | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Pérez Zeledón municipal emergency committee (CME), Nauyaca and river-tour operators, ICT	**15 Oct 2026, before the December–April high season**	**~$1–2k for radios (estimate, no quote)**	**A third flash-flood death at Nauyaca this year**

Stop all river crossings and all waterfall and pool access when rain is observed, or radar shows storm cells, over the Río Barú headwaters upstream of the Nauyaca / San Salvador sector. Volunteer observers higher up the catchment report by radio or WhatsApp. On Sunday 30 August 2026, around midday, a flash flood hit a group of 35–40 people. The guide died and two people were swept away. The Red Cross treated 35 people, and firefighters took 21 across five river crossings. The Tico Times reports that this was the second such death in Barú this year, after a US tourist was swept away at Nauyaca Falls in June. The June case was not independently verified. All of this happened during a 25% rainfall deficit: **a drought year does not mean no floods.**

* Suggested formulation: check with a health worker before sharing.

No primary evidence was found that stop rules triggered by upstream observers or rainfall reduce flash-flood deaths. The case rests on two deaths three months apart at the same site. The legal route is also unverified. The emergency law (Ley 8488) gives the committee a coordinating role, not licensing power, and ICT's tourism declaration is voluntary. Adopt the rule as a voluntary operator protocol now. Make it binding through a municipal regulation only once someone confirms that route exists.

2. Put every Barú water system on a dry-season plan by 15 November, starting with who supplies Platanillo

PARTIALLY CONFIRMED | EMERGENCY · HISTORIAN · FRONTLINE · SKEPTIC | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Barú community water boards (ASADAs); AyA Región Brunca (the Tinamaste system); CME	Weekly source readings from now; plan by 15 Nov 2026	Within operating budgets (no published figure)	Unplanned outages at dry-season peak demand; rationing that hits the poorest first

AyA, the national water utility, lists five community water boards (ASADAs) in Barú with delegation agreements: San Juan de Dios–Tres Piedras–La Reina, Las Tumbas, La Florida, San Salvador, and Chontales. A sixth, Alto San Juan, may or may not be in Barú. **Platanillo, the district centre, has no community water board on that list, and its supplier is unknown.** Tinamaste is served directly by AyA and was already rationed nightly (9 pm to 5 am) from 4 to 17 May 2026. For each system: measure source flows weekly, publish a rationing schedule that guarantees each household a minimum volume, and agree the source levels at which the emergency committee asks AyA for tanker trucks. The CNE put Pérez Zeledón on yellow alert on 18 September 2026, with Pacific-south rainfall 25% below normal from January to August.

* Suggested formulation: check with a health worker before sharing.

Corrections: the widely quoted "74 communities rationed" comes from a federation of community water systems, and it covers Guanacaste, Puntarenas, and Alajuela, not Pérez Zeledón. The 2023 figure of about 53,000 rationed residents covers four cantons combined. Do not count on AyA tankers: national demand is concentrated in the north and central Pacific, so treat them as a fallback.

3. Add covered-storage checks to the clinic's existing home visits, from the first rationing through June 2027

PARTIALLY CONFIRMED | PUBLIC HEALTH · HISTORIAN · SKEPTIC · FRONTLINE | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
EBAIS Barú (Platanillo) and its home-visit workers (ATAPs), with the Ministry of Health vector programme and the water boards	**Start with the first rationing; run to Jun 2027**	**Within existing home visits**	**A dengue surge in water-storage containers, and again when the rains return**

Every rationing notice should come with advice to lid or screen stored water and scrub containers weekly. In the Camino Verde cluster RCT (Andersson et al., *BMJ* 2015, Managua and Guerrero), community mobilisation cut dengue infection in children by 29.5% (95% CI 3.8–55.3) and reported illness by 24.7%. A Nicaragua-only secondary analysis found fewer larvae and pupae even in households with irregular water supply (house-index odds ratio 0.6). The local burden is real: the Ministry of Health declared a Pérez Zeledón outbreak in 2020 (215 cases by 23 May), and Brunca had the country's highest rate in 2024, the year after El Niño (885 per 100,000 by week 33). 2025 was quiet: 43 cases by week 27.

* Suggested formulation: check with a health worker before sharing.

Both trials ran in urban and peri-urban settings, so transfer to a dispersed rural valley is plausible but untested. The Nicaraguan result applies to larvae counts, not infection. Run through existing clinic visits, this costs little. As a standalone funded campaign it was not cost-effective (see Contraindicated).

4. Agree a dengue warning-sign referral route from Platanillo to Escalante Pradilla, including who drives

PARTIALLY CONFIRMED | PUBLIC HEALTH | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
EBAIS Barú, CCSS health area, Hospital Dr. Fernando Escalante Pradilla (San Isidro); Cruz Roja Pérez Zeledón for transport	1 Dec 2026	Staff time	Deaths from late recognition of severe dengue

Most dengue deaths come from warning signs being recognised late. The clinic and the hospital should agree the WHO/PAHO warning-sign triage, who moves the patient, and how, with a fallback if Ruta 243 is cut. The 2020 outbreak means more of this season's infections will be second infections, which carry a higher risk of severe disease.*

This rests on standard clinical guidance, not a local trial. The road distance from Platanillo to San Isidro (roughly 20–25 km on Ruta 243) was not verified.

5. Register farm and tourism workers with the social-benefits system (SINIRUBE) before anything goes wrong

CONFIRMED | ECONOMIST · FUNDER | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Volunteers with the clinic and churches; IMAS	31 Oct 2026	Volunteer and staff time	Emergency cash arriving late, or not at all

IMAS, the social-welfare institute, pays emergency transfers on an incident report from the CNE, the municipal committee, the fire service, the Red Cross, the CCSS, or the Ministry of Health. It needs no emergency decree, but it pays only households already registered in SINIRUBE. This is the only money in Costa Rica that reaches people without a decree, and a decree is unlikely while Pérez Zeledón stays

* Suggested formulation: check with a health worker before sharing.

on yellow. Guides, cattle farmers, and seasonal workers carry this season's risk uninsured.

IMAS publishes no transfer amounts. Its Brunca regional office was not confirmed.

6. Get farms and builders to comply with the existing heat-stress decree

PARTIALLY CONFIRMED | PUBLIC HEALTH · FUNDER | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
MTSS / CSO inspectors, employers	Oct–Nov 2026	Inspector time; water, shade, and rest are an employer cost	Acute kidney injury and heat illness in outdoor workers

Decreto 39147-S-TSS (La Gaceta No. 182, 18 September 2015) already requires employers of outdoor workers to provide hydration, shade, rest, and protection. At a Nicaraguan sugar mill (Hansson, Glaser et al., *OEM* 2025), a water, rest, and shade programme cut new kidney injury among burned-cane cutters from 21% to 1%, and diagnosed acute kidney injury from 20 to 8 per 1,000 worker-months.

That was an observational cohort of cane cutters, a harsher job than Barú's farm and construction work. Costa Rica's excess of chronic kidney disease of unknown cause is concentrated in Guanacaste, not Brunca, so the local risk is inferred from the mechanism. Cite it as cohort evidence, not a trial.

7. Stock 72 hours of supplies at Platanillo and list who would need moving if Ruta 243 is cut; hold this through November, then plan for 2027

UNVERIFIED | EMERGENCY · HISTORIAN | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
CME, Cruz Roja Pérez Zeledón, EBAIS Barú	31 Oct 2026	~₡3M (~$6k), a lens estimate with no quote	Deaths among dialysis patients, pregnant women, and bedridden residents while the road is closed

* Suggested formulation: check with a health worker before sharing.

Ruta 243 is Barú's lifeline. Tropical Storm Nate (October 2017; 11 dead nationally, severe road damage) cut Pérez Zeledón off, and Eta (November 2020) closed the San Isidro–Dominical road for culvert repairs. Keep food, water, first aid, and a register of residents who would need evacuating at Platanillo.*

NOAA's ENSO index shows that none of the region's big storms came in an El Niño year: César (1996) was neutral, Alma (2008) La Niña, Nate (2017) neutral turning La Niña, and Eta (2020) La Niña. So hold this readiness through the November rains, then plan for the rebound in 2027. The collapse of the Río La Palma bridge in 2017 and the March 2026 Pacuar bridge closure were not verified. The Red Cross flood protocol does not cover this area (see Funding).

8. Sign a one-page mutual-aid agreement with Osa's emergency committee covering the Dominical side of the river

UNVERIFIED · EMERGENCY · FRONTLINE · ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Pérez Zeledón and Osa CMEs	31 Oct 2026	Zero	A response gap at the canton line

The canton line runs through the valley, so people who live one community get notices from two municipalities. Rescues and evacuations up Ruta 243 cross that line. Agree who responds where, and pass river warnings to the Guardavidas Costa Ballena lifeguards, who talk to surfers and tourists every day.

No precedent was checked. Whether each committee has capacity for this is unverified.

9. Before any new water-availability letter, ask AyA to check the source can supply it

UNVERIFIED · SKEPTIC · ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Barú ASADAs; AyA decides any freeze	From now through the dry season	Forgone connection fees (unquantified)	New lots and villas drawing on springs as they fall

In Guanacaste the pattern was that availability letters were issued first and frozen only after the taps ran dry. The pressure in Barú would come from ridge lots sold toward Dominical.

* Suggested formulation: check with a health worker before sharing.

This transfers weakly. Barú is an inland valley of small rural systems, not a beach resort town. The Guanacaste precedents were not verified to a primary source in this run, and whether a community water board can suspend letters on its own authority is unconfirmed.

Deprioritised: file an ASADA tariff petition with ARESEP now, and a municipal emergency-water budget line triggered by an orange alert

UNVERIFIED ECONOMIST

A tariff petition does nothing for water supply over the December–April dry season; revisit it after April 2027 unless an ASADA cannot pay for emergency works. The municipal 2026 budget (₡14,725M) and the Law 8488 duty to budget for risk could not be confirmed to a primary source. An orange-alert trigger would not fire today because Pérez Zeledón is on yellow. The one pre-arranged national facility is the World Bank's US$160M contingent credit line for disasters (Cat DDO). It needs a formal emergency declaration, and its listed closing date is 24 March 2026 while its status still shows active. Ask the finance ministry (Hacienda) whether it was renewed.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	CME Pérez Zeledón	CME Osa	Tour operators & ICT	Volunteers & churches	Barú ASADAs	AyA Región Brunca	EBAIS Barú (ATAPs)	Ministry of Health	CCSS & Escalante Pradilla hospital	Cruz Roja Pérez Zeledón	IMAS	MTSS / CSO inspectors	Employers (farms & builders)	Guardavidas Costa Ballena
1. Flash-flood stop rule for Nauyaca and Río Barú	**A**		**R**	**R**						**I**				
2. Dry-season plan for every water system	**R**				**A/R**	**R**	**I**							
3. Covered-storage checks on clinic home visits					**R**		**A/R**	**R**						
4. Dengue referral route to Escalante Pradilla							**R**		**A/R**	**R**				
5. Register workers with SINIRUBE				**R**			**R**				**A**			
6. Enforce the existing heat-stress decree								**C**				**A/R**	**R**	
7. 72-hour stock and evacuation list, Platanillo	**A**						**R**			**R**				
8. Mutual-aid agreement with Osa's committee	**A/R**	**R**												**I**
9. AyA source check before availability letters					**R**	**A/R**								

What the chart shows. **The Pérez Zeledón emergency committee is accountable for three rows (1, 7, and 8) and does work on a fourth.** Under Ley 8488 its role is to coordinate, not to license, so the stop rule in row 1 stays a voluntary operator protocol until someone confirms a municipal regulation route. **EBAIS Barú does the work on four rows (3, 4, 5, and 7).** It is one clinic, so its staff time is the plan's main dependency. **Row 2 has an owner for every water system except the district centre.** Platanillo has no community water board on AyA's list and its supplier is unknown, so nobody is yet in a position to write its plan. **Volunteers carry an R on rows 1 and 5 with almost no money behind them.** The only funded piece is $1,500 in radios for the upstream observers, and SINIRUBE registration runs on volunteer time.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs. Nothing pre-arranged covers this place: Costa Rica has no drought framework on the Anticipation Hub, is not a CCRIF member, and the Red Cross flood protocol (EAP2023CR02, active to 31 Oct 2028) targets only the Tusubres, Parrita, and Damas basins in the Central Pacific.

* Suggested formulation: check with a health worker before sharing.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
$25,400, conditional on Cruz Roja accepting an earmark and a Barú ASADA being named	**$32,900**	**~$72,700 (provisional)**	**Who supplies Platanillo, and whether Cruz Roja Pérez Zeledón will hold a fund tied to a trigger**

Floor = $15,000 Cruz Roja fund + $10,400 tanks (40 × $260). Target = floor + $6,000 worker kits + $1,500 radios. The ceiling is the lens's $80,210 minus the $7,470 standalone container campaign that verification removed. Its other parts (up to $25k in the Cruz Roja fund, 80 tanks, 300 worker kits) were not independently checked. The target fell from the lens's $40,370 to $32,900, a drop of 18.5%, which is under the one-third threshold.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Cruz Roja Costarricense, Pérez Zeledón committee	Heat and water response: water delivery, heat-stroke care, transport for action 4. Flood evacuation removed, since that belongs to the national flood protocol	$15,000	Pre-agreed contingency fund, released if the CNE raises Pérez Zeledón to orange	Money released on a trigger arrives before the harm; nobody else holds pre-arranged money here. No cost-per-life figure exists	Partially confirmed. The committee is real and staffs Dominical beach; whether it accepts funds tied to a trigger is unconfirmed until it says so in writing
A Barú ASADA (to be named: San Salvador, Las Tumbas, La Florida, Chontales, or San Juan de Dios)	40 sealed, screened 1,100 L tanks as outage buffers for upland households (action 2)	$10,400	Pre-positioned stock	EPA lists the 1,100 L tricapa tank at ₡97,450 (~$193). $260 fitted includes about $67 for transport and fitting. The 40-household count is an assumption	Partially confirmed. Price confirmed; recipient changed away from the Escaleras road association (see below)

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Barú farms and contractors	Water, shade, and rest kits for about 150 outdoor workers (action 6)	$6,000	Direct grant	$40 per kit is a lens estimate with no evidence behind it. The programme cited (Adelante, Nicaragua) cost about $2,237 per worker and included clinical staff, and returned $1.02 per $1 averaged over 2017–22	Unverified cost basis. Get a real kit quote first, and fund Barú workers only, not Dominical
CME with upstream volunteers	Radios for the flash-flood stop rule (action 1)	$1,500	Direct grant	No quote	Unverified

The funder's one thing: here, what is scarce is organisations able to spend money, not money itself, and the district's water in Tinamaste is AyA's legal duty. Don't pay for a state utility's job. Fund what AyA won't reach — upland households, outdoor workers, and a fund that pays out on a trigger before anything is declared. The first step costs nothing: one phone call to Cruz Roja Pérez Zeledón to ask whether it will hold $15,000 on those terms.

DO NOT DO THIS — CONTRAINDICATED

Advice that circulates for this hazard and would cost lives here.

Drill new wells or pump existing ones harder to get through the drought

CONTRAINDICATED

The documented failure is in Guanacaste: overpumped coastal wells turned the Huacas-Tamarindo and Brasilito-Potrero aquifers salty, and some had to be abandoned. Barú's rural systems draw mainly on springs and surface sources, so the saltwater risk does not apply. The underlying mistake does: adding users to sources that are already falling.

* Suggested formulation: check with a health worker before sharing.

Instead: protect the spring recharge areas, add no new users to declining sources, and drill no well without a permit from SENARA, the national groundwater agency.

Route household tanks through the Escaleras road association, or treat rain tanks as dry-season supply

CONTRAINDICATED

The Asociación Vecinos de Escaleras is real, but it is a road association of uncertain canton with no record of water spending. Rain tanks fail for a simpler reason: the dry season runs December to April, exactly when water is short, so a tank filled in October is empty long before the drought ends.

Instead: buy sealed, screened tanks as buffers against network outages, fill them from the network, and distribute them through a named Barú ASADA.

Fund a standalone container-lid campaign at $7.47 a head

CONTRAINDICATED

The cost-effectiveness study of Camino Verde (*IJID*, PubMed 32179138) found it cost $29,196 per DALY averted in Nicaragua, 16.9 times GDP per capita. The authors concluded it was not cost-effective there. It was community mobilisation, not a lid handout.

Instead: carry the covered-storage message through the clinic's existing home visits and every rationing notice (action 3), at close to zero marginal cost.

"Don't use fans in the heat"

CONTRAINDICATED AS GENERIC ADVICE

Meade et al. (*Lancet Planetary Health* 2024) conclude that health agencies should advise against fan use above 35°C, especially for people who sweat less, such as adults over 65. At about 430 m in a humid valley, Barú rarely approaches that, so telling people to avoid fans altogether removes cheap relief.

* Suggested formulation: check with a health worker before sharing.

Instead: use fans and wet your skin below 35°C; above that, go somewhere cooler. The real danger is a power cut, so check on older residents when the power fails. *(Threshold corrected 2026-10-02 against the paper's abstract; the earlier 38–39°C figure was wrong.)**

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1996 (neutral)	Hurricane César	At least 10 dead and 43 missing nationally on the first night; Pérez Zeledón among the worst-hit cantons	Not documented for Barú
2008 (La Niña)	Tropical Storm Alma, Rivas de Pérez Zeledón	Rivas devastated	Roads rebuilt and the river channelled. 192 damaged families never received the state aid promised; the ₡350M earmarked "was never seen"
2017 (neutral → La Niña)	Tropical Storm Nate	11 dead nationally; road to Pérez Zeledón cut	Not documented
2020 (La Niña)	Hurricane Eta; dengue outbreak	Ruta 243 culvert failure; 215 dengue cases in the canton by 23 May	Outbreak declared
2023 (El Niño)	Drought rationing	Pérez Zeledón among four cantons with ~53,000 residents rationed (March)	Nothing documented for Barú
2024 (post-El Niño)	Brunca dengue peak	885 per 100,000 by week 33, the highest rate in the country	Not documented

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
2026 (strong El Niño)	Tinamaste rationing (4–17 May); tourist swept away at Nauyaca (June, not independently verified); flash flood at Nauyaca (30 Aug); CNE yellow alert (18 Sep)	At least 1 dead in August, a second reported in June	Nothing yet: no stop rule exists

The recurrence nobody has noticed: in this canton, damaging storms have come in the neutral or La Niña years that follow drought, not during El Niño itself. Nate (2017) followed the 2015–16 drought, Eta (2020) followed 2019, and the November 2024 national emergency declaration, which added Pérez Zeledón, followed the 2023–24 El Niño. Plan for water now and for landslides in the second half of 2027, when ground baked this year meets the rebound rains on hillsides built up since the Costanera Sur opened. The ENSO phase of each event is read from NOAA's index. The link between drought and the storm that follows it is a pattern in the record, not a demonstrated mechanism.

WHERE THE EXPERTS DISAGREE

UNRESOLVED

- **Fund household tanks** (Philanthropist) versus **stored water breeds dengue, and rain tanks are empty by January** (Skeptic). Resolved in part: sealed, screened buffer tanks filled from the network, paired with action 3. What would settle it: household larvae counts in tank-owning homes during rationing.
- **Make the stop rule a licence condition** (Economist) versus **the law gives the emergency committee coordination, not licensing** (verifier). What would settle it: a municipal legal opinion on whether a business licence (patente) can carry a safety condition.
- **Heat is a secondary hazard here** (most lenses) versus **outdoor workers face kidney injury** (Public health). The kidney evidence comes from Guanacaste and Nicaraguan cane fields. What would settle it: kidney-marker screening of Barú farm workers.

WHAT NOBODY COVERED

BLIND SPOT

The people dying in Barú this year are not in the census. Both 2026 deaths were at a commercial waterfall site, a guide and a visitor, while the lenses researched the 2,897 residents. Nobody examined the operators' incentives: how many visitors Nauyaca takes in a day, who carries liability, and whether closing on a rainy morning costs a day's revenue. Those questions decide whether action 1 happens. A second gap is livelihoods: none of the lenses priced drought losses for Barú's coffee, cattle, and fruit farmers, the district's actual economy, or checked fire risk on dry pasture.

* Suggested formulation: check with a health worker before sharing.

HOW TO READ THIS

METHOD

The panel is author-built. Seven simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people. They are not neighbourhood figures. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The location was requested as "Barú, Costa Rica" and scoped as Barú district, Pérez Zeledón canton, San José province. The first brief wrongly included Dominical, Dominicalito, and Escaleras; three lenses and a verifier corrected this, and the report is centred on the district itself. Climate Impact Lab impact region, as the report records it: not resolved. Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Five verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak).

* Suggested formulation: check with a health worker before sharing.

Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.

- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not consulted. Several agents exhausted their search budget before reaching it, and in the neighbouring Uvita run its downloadable dataset sat behind a sign-up form. No action here is marked as having or lacking a precedent there.
- **Health wording.** Passages where a wording error could hurt someone (dengue warning signs and referral, fans and cooling, and emergency supplies) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Run-specific gaps: the impact region is unresolved; several verifiers ran out of search budget; the Guanacaste availability-letter precedents, the Law 8488 text, the municipal budget document, the INS parametric product, and the ARESEP timeline were not reached; Platanillo's water supplier is unknown; and the June 2026 Nauyaca death rests on one news report.

References · Flash flood: Observador, Teletica (confirmed); Tico Times (June death unverified) · Geography: Barú de Pérez Zeledón (confirmed) · Water: AyA ASADA certification, Tinamaste rationing, CNE alert, CONAFLU rationing, 2023 rationing (confirmed, corrected) · Health: Camino Verde, BMJ 2015, irregular-supply analysis, Camino Verde cost-effectiveness, 2020 outbreak, Ministerio de Salud bulletins 33-2024 and 26-2025, Decreto 39147, Adelante, OEM 2025, Meade et al. 2024 (confirmed, corrected) · Finance: Cruz Roja flood EAP, CCRIF members, IMAS emergencies, World Bank Cat DDO P179861 (confirmed, corrected); municipal budget and INS parametric product (unverified) · Funding: Cruz Roja PZ at Dominical, EPA tank prices, Asociación Vecinos de Escaleras · History: César, Alma, Nate, Eta, 2024 emergency declaration, NOAA ONI (lens-sourced; ENSO phases verified)
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory
Generated 30 September 2026 for Barú, Pérez Zeledón, Costa Rica. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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